

PROJECT PROPOSAL & TECHNICAL APPROACH

AI Customer Support Chatbot Platform

Prepared for: CODAFRIQA

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Week 1 Supervisor Presentation Package

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1. Executive Summary

The primary goal of this project is to build a multi-tenant, secure customer service platform that combines automated, grounded AI responses with live human-agent handoffs and ticketing. The platform ensures that AI responses are strictly grounded in an organization's approved knowledge base using Retrieval-Augmented Generation (RAG).

2. Core System Architecture

2.1 Frontend Layer

- Customer Chat Widget — embedded, lightweight, real-time interface indicating status (AI, Waiting, Human Agent).
- Staff Web Portal — responsive dashboards for Administrators, Support Managers, Agents, Knowledge Editors, and Analysts.

2.2 Backend API Layer

- RESTful Services & WebSocket Gateway — secure endpoints, business logic, authentication, and real-time messaging.
- AI Abstraction Layer — a decoupled service connecting the application to LLM providers and embedding models without hardcoding vendor dependencies.

2.3 Data & Storage Layer

- Relational Database — stores organizations, users, roles, customer profiles, tickets, audit logs, and message histories.
- Vector Search / Index — stores knowledge article chunk embeddings for fast RAG retrieval.

3. System Workflows

3.1 Self-Service AI Chat & Fallback Workflow

The diagram below traces a customer message from the chat widget through knowledge base retrieval, confidence scoring, and either a grounded AI response or escalation to a human agent.

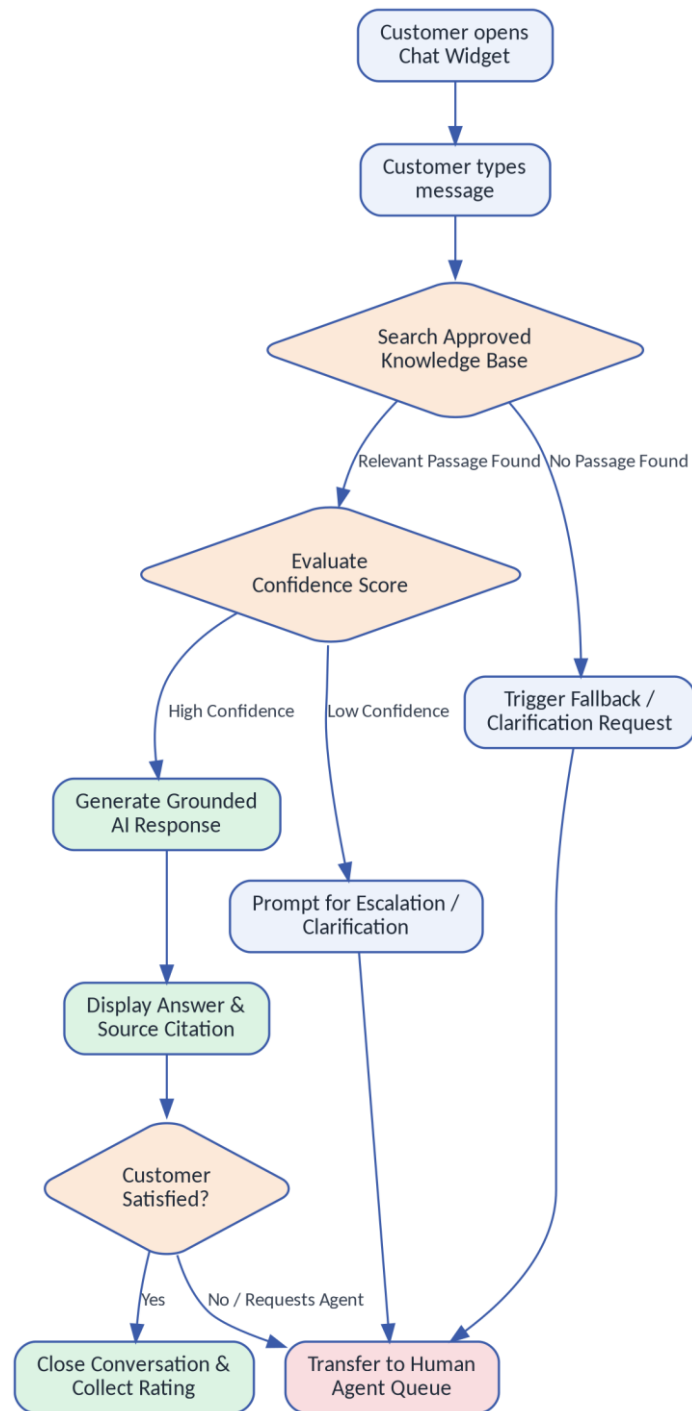


Figure 1 — Self-Service AI Chat & Fallback Workflow

3.2 Live Agent Handoff & Ticket Escalation

Once a conversation is escalated, it is queued, summarized by AI, and picked up by an available agent, who either resolves it directly or opens a tracked support ticket.

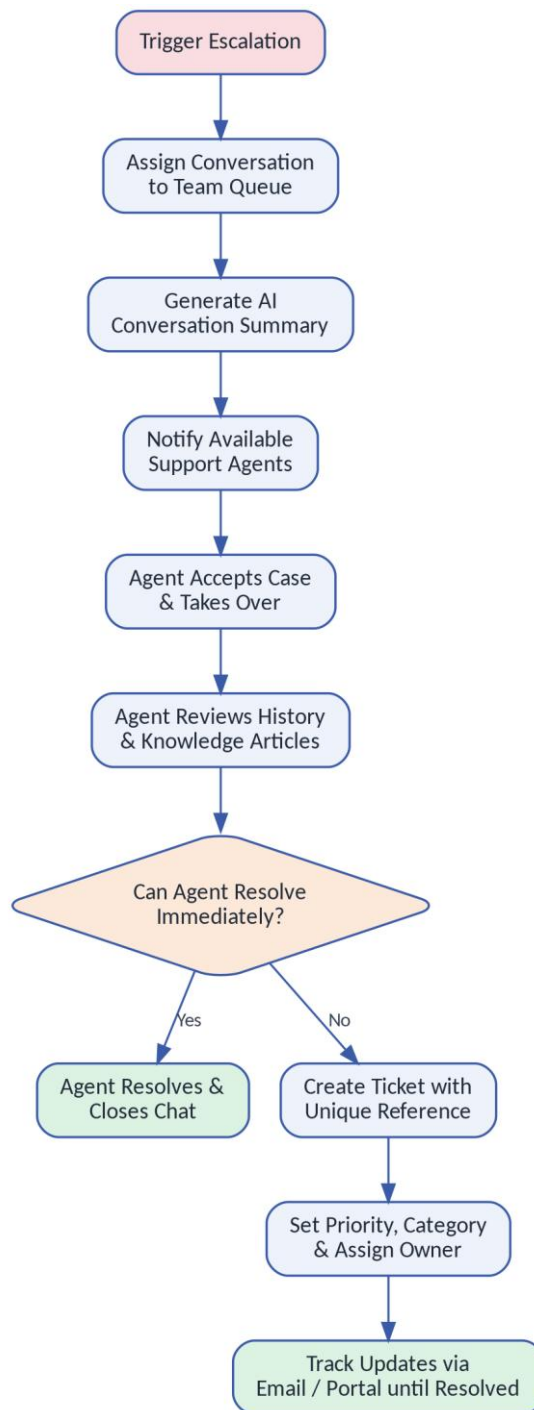


Figure 2 — Live Agent Handoff & Ticket Escalation Workflow

4. Role & Permission Matrix

Access is scoped by role to enforce least-privilege principles across the platform.

Role	Core Responsibilities & Access Permissions
System Administrator	Full platform control, organization management, global settings, API secrets, and audit logs.
Support Manager	Team queues, case assignments, service metrics, agent performance, and reporting.
Support Agent	Agent workspace, real-time chats, ticket updates, internal notes, and customer timeline.
Knowledge Editor	Create, edit, publish, version, archive, and index knowledge base articles.
Analyst / Viewer	View operational dashboards; export CSV/PDF reports without editing data.
Customer / Visitor	Initiate chat, receive grounded AI help, request an agent, submit feedback, and view own tickets.

5. AI Safety & Groundedness Rules

RAG Flow: Customer Query → Vector Search on Published Articles → Context Insertion → LLM Generation → Answer with References.

Hallucination Control: If no passage meets the similarity threshold, the model is instructed to state that the answer is unavailable and to offer human escalation.

Prompt Protection: Input validation and system prompt boundaries prevent prompt injection or cross-tenant data leaks.

6. Database Schema Draft (Core Entities)

```
[Organization] 1 --- * [User] 1 --- * [Team]
[Organization] 1 --- * [Customer] 1 --- * [Conversation] 1 --- * [Message]
[Conversation] 1 --- 0..1 [Ticket]
[Organization] 1 --- * [KnowledgeArticle] 1 --- * [KnowledgeChunk]
[User] 1 --- * [AuditEvent]
```

Key entities: Organization, User, Role, Customer, Conversation, Message, AgentAssignment, Ticket, KnowledgeArticle, KnowledgeChunk, Feedback, AuditEvent.

7. Risk Register & Mitigations

Risk	Mitigation
Unintended AI hallucination or false guidance.	Strict RAG retrieval enforcement; safe fallback rules when the similarity score is low.
Cross-tenant data leakage.	Enforce organization_id scoping at the backend controller and database query levels.
Committed API keys / secrets in Git.	Use .env templates; store secrets outside source code, in compliance with CODAFRIQA policy.

8. Suggested Presentation Plan

1. Introduction — state your name, student ID, and present the core business problem.
2. Architecture & Flow — show the two flowcharts above and walk through the RAG → Agent Handoff workflow.
3. Role Matrix & Schema — walk through how security and data isolation are enforced.
4. Milestone Commitment — confirm readiness for Week 2 (System Foundation & Access Control) once approved.